CURRICULUM BROCHURE

Solution Architect 120-Day Accelerator

A detailed path from implementation strength to architecture readiness — spanning Java, Spring, AWS, system design, security, resilience, capstone delivery, and interview storytelling.

Architecture. Cloud. Security. Delivery.

- 120 Days
- 18 Weeks
- 5 Tracks
- 1 Capstone

What this brochure packages

- An 18-week curriculum broken into weekly execution blocks
- Architecture-centered deliverables: HLDs, ADRs, runbooks, and design talks
- A capstone path that converts study into portfolio-level proof
- A disciplined blend of DSA maintenance, engineering depth, cloud delivery, and system design

Best fit

Senior backend engineers

Cloud-native developers

Lead engineers transitioning to SA

Application architect aspirants

Built to help you learn the concepts — and prove them with artifacts that hiring managers and stakeholders can actually assess.

Custom edition based on the detailed 120-day roadmap.

Program snapshot

How the 120-day path is structured, who it serves, and why the weekly blend works.

Why this roadmap works

This is not a passive reading plan. Each week mixes five lanes: DSA maintenance, Java/Spring delivery, AWS/DevOps execution, frontend support, and architecture/design. The result is a curriculum that keeps coding fluency alive while forcing you to think in systems, trade-offs, resilience, cost, security, and stakeholder communication.

Finish with proof

- Architecture diagrams and ADRs
- Deployable cloud application
- Reliability runbook and dashboards
- Security baseline and threat model
- Capstone demo and portfolio narrative

Five-track weekly model

DSA | Java / Spring | AWS / DevOps | Frontend | Architecture / Design

Suggested rhythm: 5 focused weekday sessions plus 2 heavier weekend blocks. Every weekend ends with a deliverable, retrospective, or milestone review.

Four execution phases

FOUNDATION — Weeks 1-4

- Cloud setup, API design, storage and data modeling
- NFRs, CAP, statelessness, caching, database trade-offs
- Early deployment and the first architecture milestone

Key deliverable: Day-30 architecture summary

APPLICATION — Weeks 5-8

- Spring Boot structure, microservices, containers, messaging
- Auth, queues, tracing, dashboards, resilience patterns
- Production-style engineering with observability

Key deliverable: Observable service + runbook

PLATFORM — Weeks 9-13

- Async design, caching, distributed reliability, security, IaC
- Release engineering, HA/DR, cost/performance and recovery
- System-level platform thinking and operational awareness

Key deliverable: Automated pipeline + tuned slice

ARCHITECT MODE — Weeks 14-18

- System design sprint, ADR writing, stakeholder communication
- Optional AI extension, capstone build, hardening, final talk
- Portfolio packaging and interview narrative

Key deliverable: Capstone + readiness review

Weeks 1-4 · Foundations

Architecture starts on Day 1: cloud setup, APIs, data, caching, and scaling decisions.

Week 1 — Days 1-7

Cloud setup, engineering baseline, and architecture mindset

DSA	Arrays, hashing, prefix sum, Kadane, interview complexity basics
Core	JVM memory model, class loading, JIT/GC, collections, HashMap internals
Cloud	AWS global infrastructure, account hardening, IAM, EC2, S3, budgets
Frontend	React mental model, JSX, props, state, reusable components
Architecture	Solution Architect role, Well-Architected pillars, NFRs, CAP, 3-tier thinking

Output Study repository, AWS account, first Java app on EC2, 3-tier architecture sketch

Week 2 — Days 8-14

Networking, API contracts, and secure application flow

DSA	Sliding window patterns, longest substring, minimum window, string checks
Core	Generics, lambdas, streams, Optional/records, logging and exceptions
Cloud	VPC, CIDR, public/private subnets, route tables, IGW, NAT, NACL vs SG
Frontend	Routing, params, forms, validation, hooks, light refactor discipline
Architecture	API resource modeling, REST constraints, idempotency, pagination, session vs token

Output VPC diagram, CRUD API contract, mini router app, auth decision notes

Week 3 — Days 15-21

Storage strategy, database foundations, and caching decisions

DSA	Two pointers, palindrome and 3Sum patterns, fast/slow pointers, string grouping
Core	Threads, locks, volatile, race conditions, ExecutorService, CompletableFuture
Cloud	S3 lifecycle, EBS, EFS, RDS engines, backups, read replicas, Multi-AZ
Frontend	Context, useReducer, Redux and Redux Toolkit fundamentals
Architecture	Cache-aside vs write-through, SQL vs NoSQL, ER modeling, normalization, indexing

Output RDS-connected demo, state-managed UI, database trade-off note

Week 4 — Days 22-28

Scale basics, database choice, and the first architecture milestone

DSA	Binary search, intervals, sorting fundamentals, merge intervals
Core	Parallel streams, virtual threads overview, comparator usage, unit testing basics
Cloud	DynamoDB, partition/sort keys, GSI/LSI, capacity modes, backup and restore
Frontend	API clients, loading and error states, pagination, environment config basics
Architecture	Load balancing, horizontal scaling, statelessness, sharding, consistency trade-offs

Output Small e-commerce architecture, DB decision matrix, Day-30 summary

Weeks 5-8 · Application Architecture

The program shifts from foundations into service design, deployment, eventing, and observability.

Week 5 — Days 29-35

Spring Boot service design and the move from modules to services

DSA	Linked lists, stacks, queues, parentheses, sliding-window max
Core	Spring Boot structure, IoC/DI, MVC, validation, profiles, externalized config
Cloud	Lambda, API Gateway, event handlers, Step Functions introduction
Frontend	State strategy, RTK Query, optimistic updates, state-heavy feature work
Architecture	Microservice decomposition, sync vs async calls, OpenAPI, contract thinking, saga intro

Output Spring CRUD API, OpenAPI draft, serverless comparison, monolith vs microservice note

Week 6 — Days 36-42

Persistence, containers, cloud deployment, and auth foundations

DSA	Heaps, top-K problems, DFS/BFS tree patterns
Core	JPA entities, relationships, transactions, pagination, MockMvc API testing
Cloud	Docker, ECR, ECS/Fargate, service deploy review, cloud deployment checklist
Frontend	Render-cycle debugging, memoization, lazy loading, code splitting, list virtualization
Architecture	Authentication vs authorization, JWT, OAuth2/OIDC, session vs token, secure config

Output Containerized Spring application deployed to AWS with security baseline

Week 7 — Days 43-49

Messaging, event-driven architecture, and operational correctness

DSA	Tree diameter, balancing, LCA, serialization and tree reasoning
Core	Connection pools, indexing/query tuning, Redis in Spring, structured logging, Testcontainers
Cloud	SQS, SNS, EventBridge, comparison matrix, DLQ and retry handling labs
Frontend	Scalable folder structure, reusable component patterns, component library starter
Architecture	Event-driven vs request-response, eventual consistency, outbox, idempotency, schema evolution

Output Cache-plus-queue service, event pipeline, async processing path, updated diagrams

Week 8 — Days 50-56

Observability, resilience, and measuring the health of a service

DSA	Graphs, BFS/DFS, connected components, topo sort, clone graph
Core	Feign, central config, service discovery patterns, Resilience4j circuit breaker and retry
Cloud	CloudWatch logs/metrics, alarms, dashboards, X-Ray tracing, autoscaling signals
Frontend	Jest, React Testing Library, API mocks, E2E basics for one full flow
Architecture	SLIs, SLOs, SLAs, timeouts, retries, circuit breakers, bulkheads, chaos thinking

Output Observable service, dashboards, runbook, reliability review

Weeks 9-12 · Scale, Security, and Platform Thinking

Here the roadmap becomes architect-heavy: async systems, resilience, security, release engineering, and IaC.

Week 9 — Days 57-63

Asynchronous design, Redis strategy, and measurable performance gains

DSA Dynamic programming basics: Fibonacci, Climbing Stairs, House Robber, coin variants

Core Async APIs, virtual threads vs pools, producer-consumer, cancellation and async error handling

Cloud Redis/ElastiCache, session patterns, TTL, eviction, hot keys, cache benchmarking

Frontend TypeScript basics, typed props/state, typed API responses, config cleanup

Architecture Cache selection, invalidation, request coalescing, cache boundaries, CDN vs API cache

Output Cache-backed async feature, latency before/after report, cache decision note

Week 10 — Days 64-70

Distributed reliability, orchestration choices, and consistency trade-offs

DSA Advanced DP and greedy crossover problems for interview maintenance

Core Circuit breaker tuning, retries, timeouts, fallbacks, rate limiting, load testing, graceful degradation

Cloud EKS concepts, pods/services/ingress, config and secrets, ECS vs EKS decision work

Frontend Component composition and reusable hook patterns to support larger applications

Architecture Distributed transactions, saga choreography vs orchestration, CQRS intro, consistency models

Output Resilience lab, rate-limiter notes, ECS-vs-EKS memo, resilient-service milestone

Week 11 — Days 71-77

Security architecture and clean system boundaries

DSA Jump Game, meeting rooms, intervals and priority-queue greedy review

Core Factory/strategy patterns, immutability, clean architecture, hexagonal architecture, module boundaries

Cloud KMS, Secrets Manager, Parameter Store, Cognito concepts, WAF, Shield, least-privilege IAM

Frontend Route-level code splitting, frontend structure, error boundaries, build optimization basics

Architecture Zero trust, OWASP Top 10, API security, mTLS, private APIs, threat modeling

Output Threat model draft, security checklist, architecture-pattern comparison

Week 12 — Days 78-84

Release engineering, infrastructure as code, and DevSecOps discipline

DSA Subsets, permutations, combination sum, backtracking, bit-manipulation refresh

Core Maven lifecycle, plugins, coverage, Sonar/Jacoco, versioning and release management

Cloud CodeBuild/CodePipeline or GitHub Actions, CloudFormation, Terraform, IaC drift and state

Frontend Build pipeline, bundle analysis, environment handling, source-map and monitoring basics

Architecture CI/CD principles, branch strategy, artifact promotion, release flow, DevSecOps checklist

Output Automated build pipeline, infra template, rollback and release notes

Weeks 13-15 · Performance, design rounds, and specialization

The curriculum now asks you to defend architecture choices in public: cost, DR, ADRs, and presentation become visible outputs.

Architect shift

You are no longer just building features. You are tuning systems, writing decisions, and presenting trade-offs clearly enough for stakeholders and interviewers to trust your judgment.

Week 13 — Days 85-91

Performance, cost awareness, global delivery, and recovery planning

DSA	Mixed medium timed sets and correction-focused review	**Frontend**	Accessibility review, performance budgets, semantic HTML, keyboard navigation essentials
Core	JFR/JMC, GC tuning flags, heap and thread dumps, pool tuning, DB/query profiling	**Architecture**	Cost vs performance, RTO/RPO, Multi-AZ vs Multi-region, final Well-Architected check
Cloud	Route 53, DNS, CloudFront, ACM, PrivateLink, cost explorer, optimization levers	**Output**	**Tuned and audited system slice with HA and cost notes**

Week 14 — Days 92-98

System design sprint: rate limiting, notifications, uploads, and payments

DSA	Design-focused mix and timed mocks to keep coding fluency alive	**Frontend**	Error logging, user analytics, feature flags, distributed tracing visibility from UI to API
Core	Rate limiter, notification system, URL shortener, file upload, and payment design slices	**Architecture**	CAP trade-offs, blue-green vs canary, ADR writing, stakeholder communication and design talk
Cloud	HA patterns, backups, DR, autoscaling, failover, queue buffering and decoupling	**Output**	**Architecture diagrams, DR notes, ADR set, and first design presentation**

Week 15 — OPTIONAL SPECIALIZATION — 105

Optional AI/RAG extension for solution patterns that need modern assistant features

DSA	Weak-area maintenance sets only; focus shifts toward specialization	**Frontend**	Chat UI skeleton, streaming response handling, conversation state and citation display patterns
Core	LLM API integration, embeddings, chunking, retrieval flow, guardrails, evaluation strategy	**Architecture**	Prompting trade-offs, hallucination controls, AI security/privacy, memory and grounding strategy
Cloud	S3 knowledge base, ingestion pipeline, async processing, rate limits, latency/cost controls	**Output**	**Optional GenAI slice, evaluation rubric, and AI design notes**

Weeks 16-18 · Capstone execution and final packaging

The last stretch turns the roadmap into proof: a deployable solution, supporting documents, and a presentation-ready narrative.

Week 16 — **Days 106-112**

Capstone launch: scope, architecture baseline, and staging readiness

DSA	Mocks and weak-area correction only; architecture execution takes priority	**Frontend**	App shell, route map, auth flow, API client, feature pages, UX baseline
Core	Capstone scope, domain model, auth service, core business service, async/event service, test plan	**Architecture**	NFRs, acceptance criteria, ADRs, deployment checklist, backlog shaping, architecture version 1
Cloud	Reference architecture selection, IAM/networking, storage, database, queue/cache, monitoring baseline	**Output**	**Capstone backlog, working auth skeleton, core API, staging-ready baseline**

Week 17 — **Days 113-119**

Capstone hardening: security, performance, recovery, and presentation readiness

DSA	Final mock sequence for confidence, not depth expansion	**Frontend**	Frontend tests, performance fixes, accessibility pass, docs, final UX polish
Core	Optimize hotspots, security hardening, failure handling, CI/CD hardening, code cleanup, rehearsal	**Architecture**	Runbook, incident scenarios, metrics dashboard, stakeholder Q&A, portfolio and resume bullets
Cloud	Autoscaling and cost controls, CloudFront/WAF/Route53, backup and DR review, deployment validation	**Output**	**Release candidate, polished demo, security review, presentation rehearsal**

Week 18 — **Day 120 and beyond**

Final assessment, presentation, and next-step acceleration plan

DSA	Final mock and pattern-sheet consolidation	**Frontend**	Frontend architecture recap focused on integration, monitoring, and delivery support
Core	Java/Spring interview notes, strengths review, and targeted follow-up topics	**Architecture**	Final Solution Architect presentation, readiness review, and the next 90-day improvement plan
Cloud	AWS review across compute, storage, networking, security, operations, and cost	**Output**	**Recorded final assessment and a continuation roadmap for interviews or promotion**

Capstone artifact pack

HLD and LLD diagrams · deployment view · sequence diagrams · ADR set · runbook · threat model · SLOs and metrics dashboard · cost and HA notes · polished demo walkthrough · resume and portfolio bullets.

Why this brochure is worth executing

A roadmap is only valuable if it leaves you with visible, defensible proof of architecture readiness.

Build the proof, not just the knowledge.

READY-TO-TRACK

This 120-day curriculum is designed to turn study time into architecture artifacts, delivery evidence, and a credible Solution Architect story.

Who this is for

- Senior software engineers
- Backend or platform engineers
- Cloud-native developers
- Leads moving toward solution ownership
- Professionals preparing for application or solution architect interviews

What you walk away with

- A capstone that is actually demoable
- Decision documents, not only code
- Cloud deployment and reliability evidence
- Architecture talk track for interviews
- A measurable readiness review

How to use it best

- Pair it with the tracker workbook
- Review every weekend
- Write one architecture artifact weekly
- Keep DSA as maintenance, not the main event
- Turn every milestone into portfolio material

What makes the path persuasive

1. It teaches cloud, software delivery, system design, and stakeholder reasoning together instead of in silos.
2. It ends with architecture artifacts that can be shown, explained, defended, and improved.
3. It creates a bridge from “I can build this” to “I can choose, justify, secure, scale, and present this.”
4. It is broad enough to build range and structured enough to keep momentum for 120 days.

Execute the full 120 days with discipline and you will not just study architecture — you will finish with a body of work that makes your transition easier to explain, evaluate, and trust.

Companion asset: tracker workbook for weekly execution, readiness scoring, and milestone review.